Engine9 Programming Language

Language for Engineering Calculations

Final Project

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Abstract

My project presents the design and implementation of Engine9, a domain-specific programming language focused on engineering calculations. The language enables users to write concise programs to compute values such as force, energy, and electrical current in a matter of seconds. The system includes a tokenizer, parser, abstract syntax tree, symbol table, and interpreter. The goal of this project is to demonstrate fundamental compiler concepts while providing a practical tool for solving real-world engineering problems.

Introduction

Programming languages are powerful tools, but many are designed for general-purpose use. Engineering students often need to perform straightforward calculations using formulas, yet they must work within the constraints of complex syntax and programming constructs.

Engine9 was developed to provide a simpler alternative that allows users to focus on solving problems rather than writing boilerplate code. The language supports variable declarations, arithmetic expressions, and built-in engineering functions.

Problem Statement

Current programming languages such as Python and Java are not optimized for simple engineering calculations. They require additional setup, such as defining functions and managing program structure, which can be unnecessary for straightforward tasks.

This creates a barrier for beginners and increases the time required to perform basic calculations.

Background and Motivation

Engineering disciplines rely heavily on mathematical formulas. Students and professionals frequently compute values such as force, energy, and electrical current. A domain-specific language tailored for these tasks can improve efficiency and usability.

Engine9 is motivated by the need for a lightweight language that reduces complexity while still demonstrating core programming and compiler principles.

Core Language Features

- Variable declaration using let
- Arithmetic operations (+, -, *, /)
- Built-in engineering functions
- Expression evaluation
- Output using print
- Symbol table for variable management

Type System

Engine9 uses a simple dynamic type system. Variables are assigned types based on their values at runtime. Supported data types include:

- Integer
- Float

Example Syntax and Semantics

Example Engine9 program:

```
let mass = 10
let accel = 9.8
let forceValue = force (mass, accel)
print forceValue
```

Semantics:

- Variables are declared using let
- Expressions are evaluated from left to right
- Built-in functions compute engineering formulas
- print outputs results to the console

Compiler Architecture

Engine9 follows a simplified compiler pipeline:

1. Lexical Analysis (Tokenizer)
2. Parsing
3. Abstract Syntax Tree (AST)
4. Symbol Table Management
5. Interpretation (Execution)

Benefits of Engine9

- Simplifies engineering calculations
- Reduces unnecessary syntax
- Demonstrates compiler concepts
- Easy to learn and use
- Focused on real-world applications

Future Work

- Add control flow (if, loops)
- Expand built-in function library
- Improve error handling
- Develop a graphical interface

Conclusion

Engine9 successfully demonstrates the design and implementation of a domain-specific programming language. It simplifies engineering calculations while showcasing key compiler concepts. The project provides both educational value and practical application potential.